

Problem Solving (MATH 1000)

Summer 2018

Thursday, 06/07/2018

Quiz 3: Section 10.1, 10.2, 10.3

Name (Print): _____

Score: _____ / 100

1. Write the following percentages as decimal values:

(a)

5 points

 0.35%

(b)

5 points

 350%

2. Write the following decimal values as percentages:

(a)

5 points

 0.35

(b)

5 points

 350

3. 15 points Suppose that you deposit \$4,000 in a savings account that pays 2% annual interest, with interest credited to the account at the end of each year. Assuming that no withdrawals are made and that interest is compounded annually, find the balance in the account after 3 years.
4. 15 points For an investment of $P = \$1,262.50$ compounded annually with a 6.75% APR, find the value of the investment after 2 years. Then, find the value of the investment after 3 and a half years.

5. 15 points Between the years 1994 and 2004, the average annual inflation rate was 2.7%. Find the salary in the year 2004 that would be equivalent to a \$35,000 salary in 1994.

HINT: If inflation *reduces* the value of your money (relative to some measurement of the average price of things you can buy with a dollar) by 2.7% per year, then your salary needs to *increase* by 2.7% per year in order for it to retain an equivalent value after adjusting for inflation.

6. Congratulations! You just a baby named George! [If you were not expecting this or thought the baby might be named something other than George, also “Surprise!”] You have decided that George needs to go to college, but must have \$40,000 in a savings account before he starts.
- (a) 15 points How much money do you need to put in a savings account that pays 3% compounded quarterly (that is, 4 times per year) so that your George can go to college on his 18th birthday?

- (b) 20 points If you put \$25,000 in a savings account that pays interest compounded monthly, what interest rate is needed so your George can go to college on his 16th birthday?

HINT: $\sqrt[192]{\frac{40,000}{25,000}} \approx 1.00245093421$